	AzureSky	
	Homogeneity & Stability	**Document no: 7.10**
	Compiled by: **Tshepo Maponya - Laboratory Supervisor**	
	Approved by: **Sifiso Nyambi - Laboratory Manager**	

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1. Purpose

This procedure describes the process for assessing and monitoring the homogeneity and stability of coal reference materials (RMs) and proficiency testing (PT) samples prepared by the lab. The procedure ensures compliance with the requirements of **ISO 13528:2020 (Annex B)**, **ISO 17034:2016 (Clauses 7.10 and 7.11)**, **ISO 17043(Clause 7.3.2)** and **SANAS F149**.

2. Scope

This procedure applies to all coal-based reference materials and PT samples produced internally by the lab. It covers the sampling, testing, statistical evaluation, and ongoing stability monitoring to ensure that all produced reference materials are homogeneous and stable over their intended use period.

3. References

- ISO 17034:2016 — *General requirements for the competence of reference material producers*
- ISO 13528:2020 — *Statistical methods for use in proficiency testing by interlaboratory comparison* (Annex B)
- ISO 1171 — *Solid mineral fuels — Determination of ash*
- SANAS F149 — *Assessment Checklist for Reference Material Producers*
- ISO/IEC 17025:2017 — *General requirements for the competence of testing laboratories*

4. Definitions

- **Homogeneity**: The degree to which a property value of a reference material is consistent among different units of the same batch.
- **Stability**: The ability of a reference material to maintain its assigned property value within the stated uncertainty over a defined time period under specified storage conditions.

- **σpt (Sigma PT)**: Standard deviation for proficiency testing derived from PT data or from expected variability of the property.
- **Check value**: Acceptance limit for between-sample variation.

5. Responsibilities

- **Laboratory Manager / Supervisor**:
  - Oversees implementation of this procedure.
  - Approves results, records, and interpretation of homogeneity and stability testing.
  - Ensures results are reviewed, recorded, and archived in compliance with ISO 17034 record control requirements.
- **Analyst**:
  - Conducts the analytical determinations as per ISO 1171.
  - Records results in the Homogeneity and Stability Assessment Worksheet.

6. Equipment and Materials

- Muffle Furnace capable of maintaining 815 ± 10 °C
- Analytical balance (calibrated, SANAS-traceable)
- Pulverizer and homogenizer (checked for cleanliness before use, efficiencies performed)
- Clean sampling scoops and sample trays
- Labeled 150 g sample bottles with seals
- Environmental monitoring device (temperature and humidity)
- Laboratory Information Management System (LIMS) or Excel-based data sheet

7. Procedure

7.1 Sample Preparation

1. Pulverize the coal material to the required fineness (-212µm).
2. Homogenize the material in a mechanical homogenizer for **4 hours**.
3. Extract at random, ten 600g portions of the material (~80 kg batch).
4. Using the **flat and heap method**, extract **150 g** aliquots from ten randomly selected portions and label them with their respective LIMS numbers.

8. Homogeneity Assessment

8.1 Analytical Testing

1. Analyze the 10 samples for **Ash content** in duplicate (R1 and R2) as per procedure AS003IW**.**
2. Record individual replicate and mean values in the **Homogeneity Assessment Worksheet**.

8.2 Statistical Evaluation (per ISO 13528: Annex B.2)

1. Calculate the **mean** ($X\bar{}$) and **range** for each sample.
2. Perform **one-way ANOVA** to determine:
    - Between-sample variance (SS_{between})
    - Within-sample variance (SS_{within})
3. Calculate (sigma PT**σ**) as:

$$\sigma_{pt} = SD_{PT} \times \bar{X}$$

4. Determine the **check value**:

$$Check\ Value = 0.3 \times \sigma_{pt}$$

5. **Decision Rule 1 (Check Value):**
   If ss between is less than the check value → The material is **homogeneous**.

9. Stability Assessment

9.1 Initial Stability Test

1. Store the same 10 bottles used for homogeneity testing in the **CRM storage room** (16–20 °C; humidity monitored).
2. After 30 days, re-analyze the same 10 bottles for **Ash content**.
3. Calculate the mean difference (Δ) between homogeneity mean and stability mean:

$$\Delta = | \bar{X}_{stability} - \bar{X}_{homogeneity} |$$

4. Compare Δ to the **check value**
   - If Δ < Check Value → Samples are **stable**.
   - If Δ > Check Value → Investigate cause and re-evaluate material.

9.2 Ongoing Stability Monitoring

1. Conduct monthly stability monitoring on the **25th of each month** for a **12-month period**.
2. Use archived bottles stored under reference conditions.
3. Analyze for ash content and compare to the **assigned value** of the CRM.

4. Criteria for ongoing stability:
    - Monthly average must be reproducible to the assigned value.
    - Any drift exceeding this limit must trigger an investigation and possible re-evaluation.
5. If stability issues ensue, the CAB documents that stability is to be re-done for the month and if it still does not meet the criteria set. These are the steps to be followed:
    - The technicians choose at random from the produced lot, 10 bottles of the CRM of interest and analyze for validation. Ash results are used to test for homogeneity and after the set time, reanalyzed to test for stability and then the rest are used for parameter re-evaluation.
    - New uncertainties will be assigned since the homogeneity and stability studies make up the uncertainty budget for value assignments.
    - The CAB may or may not choose to re-issue this lot/batch for PT distribution so it can also reassign the values in their entirety.
    - When the CAB follows this route, it will then be prompted to communicate these findings to its customers for the affected CRM and to issue amendments to the already produced certificates.

10. Records

All results, calculations, and conclusions shall be documented in controlled templates:

- Stability Monitoring Excel Spreadsheet

All records are to be retained for **10 years** or the validity period of the CRM batch, whichever is longer.

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12. References for Compliance

- ISO 13528 Annex B.2 (Testing for sufficient homogeneity)
- ISO 13528 Annex B.3 (Testing for stability)
- ISO 17034 Clause 7.10 (Homogeneity)
- ISO 17034 Clause 7.11 (Stability)
- SANAS F149 (Sections 7.10 and 7.11)